



FX PB Collateral

Automating FX Prime Brokerage Monitoring

Replacing tedious manual portal extractions with an automated end-to-end data pipeline & Power BI analytics dashboard



The Business Problem & Enterprise Impact

Understanding the manual bottlenecks in daily FX Prime Brokerage collateral and limits reporting.



Manual Process Challenge

1

Portal Authentication

Operations team had to manually log into the Prime Brokerage (PB) portal every morning, navigate through reporting sub-menus, and await slow report loads.

2

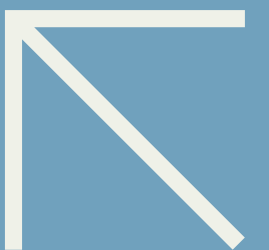
Manual Screenshotting

Gather relevant quantitative and qualitative data from internal and external sources

3

Teams Pasting & Delays

Clean incomplete records, remove duplicates, and ensure accuracy to maintain data integrity



Implemented Solution

Automated Ingestion Pipeline



Automated Email Delivery

PB portal configured to auto-distribute daily CSV files



Power BI Dataflows

Connects to SharePoint, cleans, aggregates, and structures the raw CSV data for downstream reporting



Power Automate Workflow

Cloud workflow automatically saves CSV files to a secured SharePoint folder



Power BI Reports

Automatically refreshed interactive reports distributed to the primary stakeholders

Operational Speed & Impact

<10s

Time to Decision
Insight

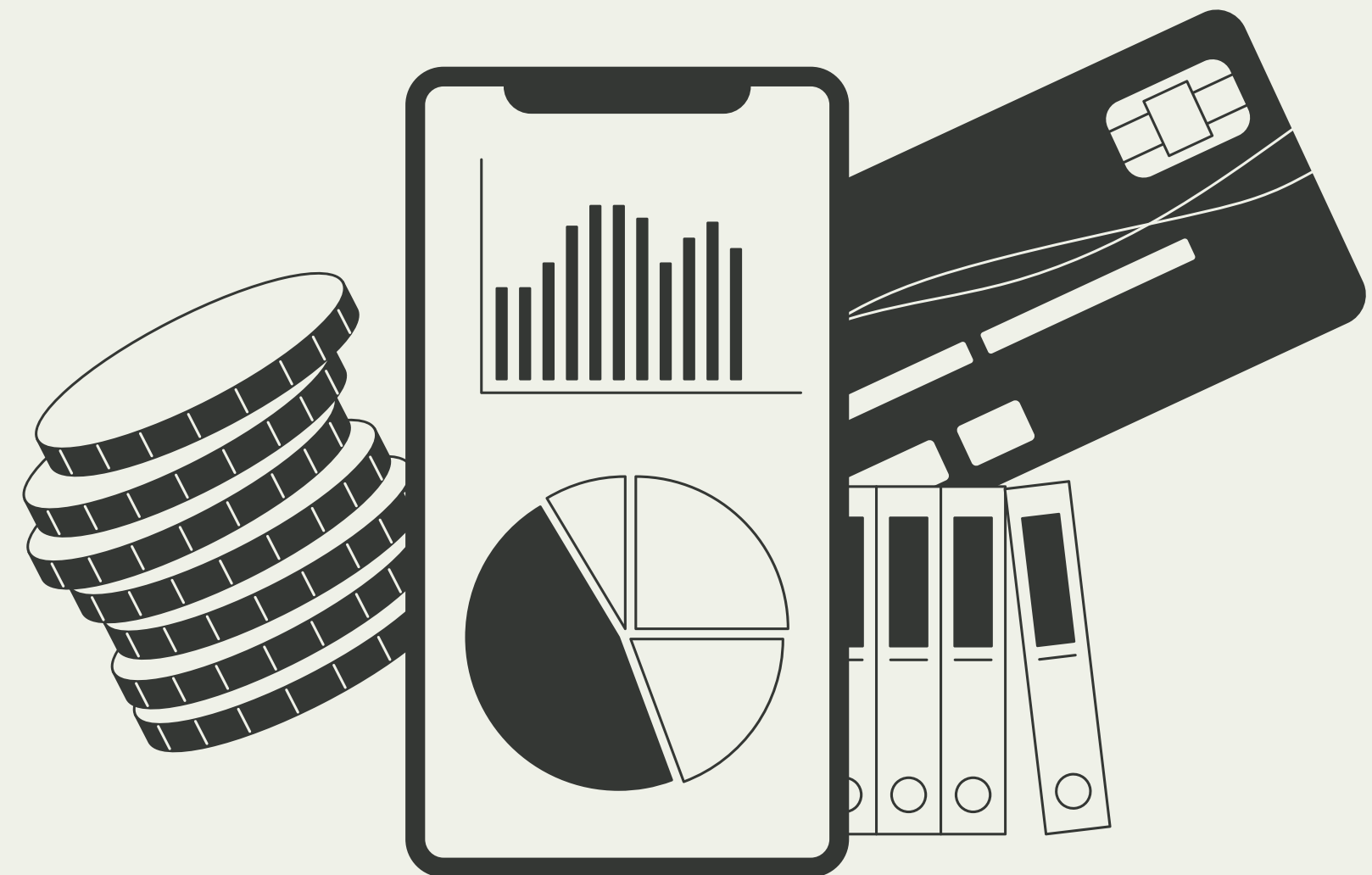
Instant Collateral Requirements Visibility

By removing manual portal logins, screenshotting, and Teams messaging, the solution compresses the daily reporting cycle from over 15 minutes down to sub-second automated dashboard access.

Stakeholders receive updated, reliable, and queryable collateral data before the market opens every morning.

Public Portfolio & GitHub Replication

Enterprise project securely recreated into an open-source demonstration





Enterprise vs Open Source



Workplace Production		GitHub Replication
Prime Brokerage portal CSV feeds	Data	Custom Python script generating synthetic FX data
Microsoft SharePoint Online	Storage	PostgreSQL on Neon Database
Power BI Dataflows to BI reports	Connection	Direct Power BI connection to Neon DB.
Power BI Report	Reporting	Power BI Report (other tools will be used)

Data Architecture



Python & Neon PostgreSQL Pipeline



Data Generation

Developed a Python script to synthesize mock data mimicking original reports



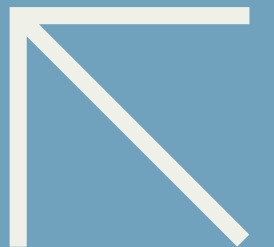
Database Integration

Structured a relational schema and automated data loads directly into a Neon PostgreSQL serverless cloud database



Power BI Reports

Connected Power BI Reports directly to Neon Database to model data, write DAX measures, and publish the dashboard



AI-Assisted Development

1

Rapid Scripting & Synthetic Logic

Leveraged Generative AI assistance to program complex Python generator scripts

2

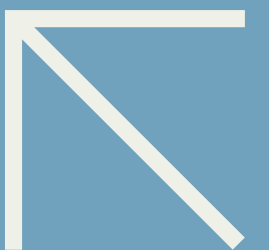
Database Schema Optimization

Leveraged AI prompts to help design PostgreSQL tables— schema layout, primary keys, and data types for Neon DB

3

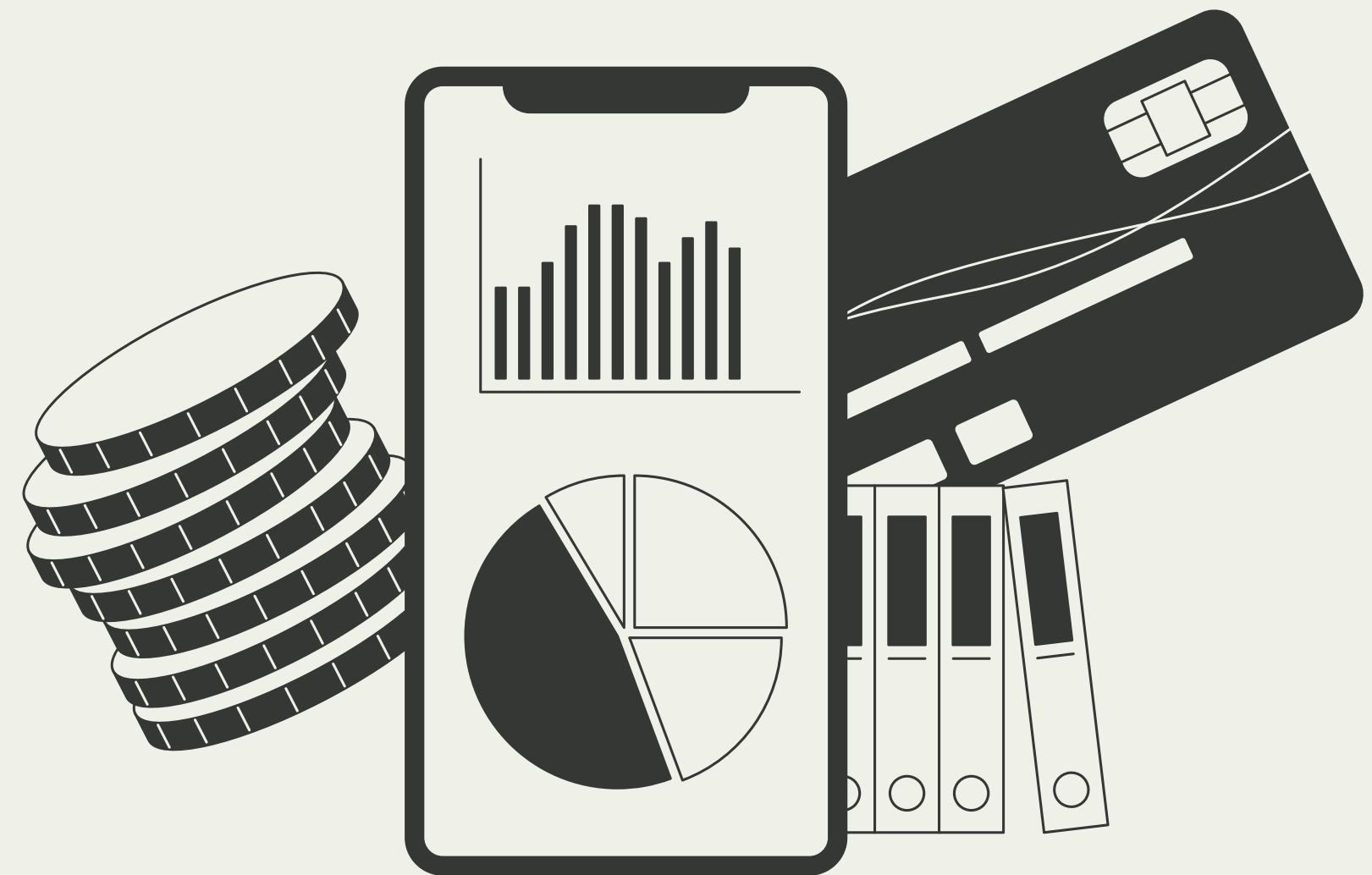
Documentation

Generated docstrings and a comprehensive GitHub README explaining setup instructions for reviewers



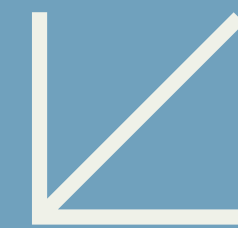
Reporting Examples

Interactive Power BI Reports





Collateral Dashboard



Collateral Dashboard

Selected Date
10/Aug/2026

Initial Margin
-5.09M
Change: ▲ 588K

Variation Margin
-46.24K
Change: ▼ 41K

Collateral Required
-5.13M
Change: ▲ 547K

Collateral Posted
6.19M
Change: ▲ 1M

Excess / Deficit
1.06M
Change: ▲ 2M

Total Portfolio (In USD)
102.93M
Change: ▼ 11M

Open Positions
45
Change: ▼ 1

☐ Latest Date
☒ 10-Aug-2026
☐ 07-Aug-2026

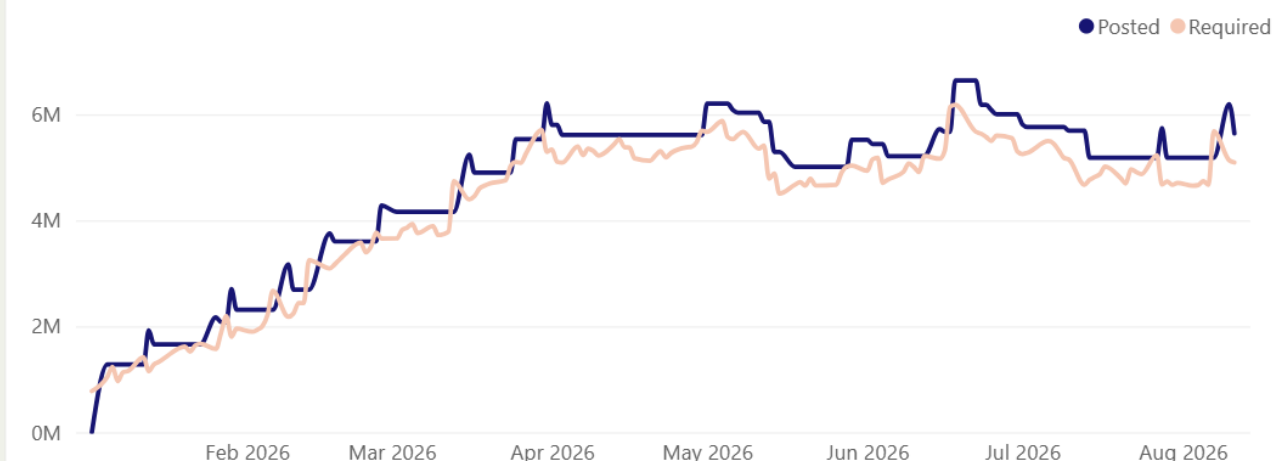
Initial Margin

IM vs 30 Day Moving Average



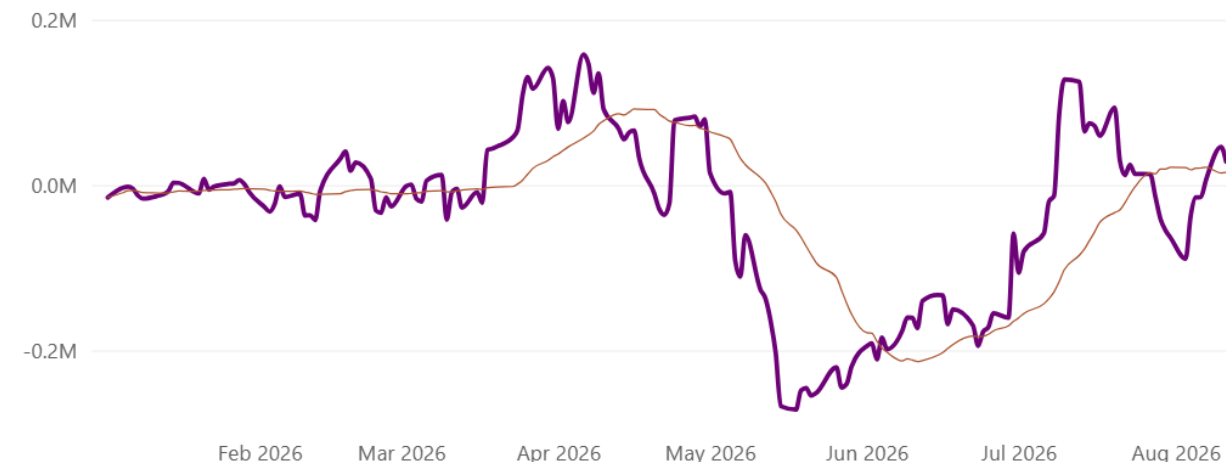
Collateral: Required vs Posted

Posted vs Required



Variation Margin

VM vs 30 Day Moving Average



Excess / Deficit

